Exercise 3

Linear Algebra — Transition Matrices and Markov Chains

Due date: June 24

Please submit both your LLM chat log and a Python notebook.

Part A

Consider the following 5×5 transition matrix T, which operates on column vectors $\mathbf{P}$ of length 5:

$$T = \begin{pmatrix} 1/2 & 1/2 & 0 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 1/2 & 0 & 1/2 \\ 0 & 0 & 0 & 1/2 & 1/2 \end{pmatrix}$$

The dynamics are given by

$$\mathbf{P}(t+1) = T\,\mathbf{P}(t), \qquad \text{with initial condition} \qquad \mathbf{P}(0) = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

1. Calculate $\mathbf{P}(2)$ and $\mathbf{P}(3)$.
2. What is special about the columns of the matrix T?
3. Write a general formula for $\mathbf{P}(t)$ as a function of T and $\mathbf{P}(0)$.
4. Calculate the eigenvalues and eigenvectors of T.
5. Calculate the determinant of T. If T is invertible, what is its inverse?
6. Assume that $\mathbf{P}(105)$ is given. Can you calculate $\mathbf{P}(104)$?
7. Can you approximate the vector $\mathbf{P}(t)$ when t is very large?
8. Would the result of question 7 change had we selected a different value for the vector $\mathbf{P}(0)$? Do you perhaps see a subtle contradiction with question 5?

Part B

Answer questions 1–8 from Part A for the following transition matrix T (with the same $\mathbf{P}(0)$):

$$T = \begin{pmatrix} 0 & 1/2 & 0 & 0 & 0 \\ 1 & 0 & 1/2 & 0 & 0 \\ 0 & 1/2 & 0 & 1/2 & 0 \\ 0 & 0 & 1/2 & 0 & 1 \\ 0 & 0 & 0 & 1/2 & 0 \end{pmatrix}$$

Part C

Can you think of a “story” — a real-world problem — that the mathematical process above may describe?

Part D

Imagine that the matrix T of Part A was 100×100 and that $\mathbf{P}(0)$ was of length 100 (with the single nonzero entry in the middle).

Plot the vector $\mathbf{P}(t)$ for $t = 0$, $t = 50$, $t = 100$, and $t = 500$.